

TABLE 3—EFFECTS OF CAPS ON MENTAL HEALTH HOSPITALIZATION: ADMISSIONS PER 10,000 PEOPLE

	2-year effect (1)	5-year effect (2)	Placebo effect (3)	Mean at baseline (4)
<i>Mental health admissions</i>				
Overall	−0.839 (0.249)	−0.951 (0.272)	−0.076 (0.066)	12.662
Long-stay	−0.464 (0.151)	−0.621 (0.155)	−0.074 (0.054)	6.035
<i>Admissions by cause</i>				
Schizophrenia	−0.602 (0.113)	−0.823 (0.122)	−0.033 (0.033)	5.359
Psychoactive substance abuse disorders	−0.039 (0.179)	0.048 (0.183)	−0.026 (0.037)	4.280
Mood disorders	−0.108 (0.091)	−0.077 (0.092)	0.011 (0.018)	1.632
Anxiety and stress- related disorders	−0.006 (0.011)	−0.003 (0.011)	−0.005 (0.004)	0.111

Notes: This table reports the average effects of CAPS on the number of mental health hospital admissions related to mental health per 10,000 people. The first panel reports results on overall mental health admissions (ICD-10 F00-F99) and long-stay hospital admissions due to mental and behavioral disorders (> 30 days). The second panel shows results by cause: schizophrenia and related disorders (F20-F29), mood disorders (F30-F39), anxiety and stress-related disorders (F40-F48), and substance abuse disorders (F10-F19). Standard errors in parentheses are computed using a municipality-level clustered bootstrap. Controls include municipality GDP per capita, PBF spending per capita, a series of indicators for age-by-gender population bins, state $\times$ year fixed effects, and linear trends of pretreatment characteristics. Pretreatment municipality characteristics included are: Theil index, poverty rate, unemployment rate, illiteracy rate, share of rural population, log of population, log of social spending, number of mental health providers, number of mental health offices, municipality area, altitude, distance to capital, temperature, and rainfall. Columns 1 and 2 report the average effect after two years and after five years, respectively. Column 3 reports the placebo effect. Column 4 shows the mean of each variable at baseline—i.e., one period before the event.

effects on well-being. To evaluate this impact, we would need a way to estimate the impact of the policy on less severe aspects of mental health, the benefits of more humane treatment compared to treatment delivered at hospital inpatient settings, and the potential externalities of deinstitutionalization to the community and patients' families.

D. Deaths of Despair

We now turn to our main measures of mental health, mortality rates (per 10,000 people). In particular, we look at deaths related to alcohol abuse, overdose, suicide, and deaths of despair. Figure 7 and Table 5 present our results. Overall, we find no statistically significant effect on mortality rates. Based on average effects, we have negative and insignificant estimates for alcohol-related deaths and deaths of despair in general. Based on the estimated standard errors, we would be able to detect effects at least as large in absolute value as 0.051 and 0.066 deaths per 10,000 people for alcohol-related deaths and deaths of despair, respectively. These correspond to 3.8 percent and 3.2 percent of the baseline values of these outcomes, respectively. As for suicides and overdoses, the estimates are small and positive.

TABLE 4—EFFECTS OF CAPS ON PUBLIC HEALTH SPENDING

	log(<i>Mental health hospital spending</i>) (1)	log(<i>Overall hospital spending</i>) (2)	log(<i>Municipality health spending</i>) (3)
2-year effect	−0.138 (0.038)	−0.004 (0.006)	0.117 (0.092)
5-year effect	−0.157 (0.032)	0.003 (0.007)	0.048 (0.082)
Placebo effect	−0.003 (0.010)	0.001 (0.002)	0.046 (0.030)
Baseline	8.95	13.969	14.609

Notes: This table reports the average effects of CAPS on the log of different health spending measures. *Mental health hospital spending* is the federal expenses with hospitalizations associated with mental health. *Overall hospital spending* is the federal expenses with hospitalizations in total. *Municipality health spending* is the health expenses incurred by local governments. Standard errors in parentheses are computed using a municipality-level clustered bootstrap. Controls include municipality GDP per capita, PBF spending per capita, a series of indicators for age-by-gender population bins, state $\times$ year fixed effects, and linear trends of pretreatment characteristics. Pretreatment municipality characteristics included are: Theil index, poverty rate, unemployment rate, illiteracy rate, share of rural population, log of population, log of social spending, number of mental health providers, number of mental health offices, municipality area, altitude, distance to capital, temperature, and rainfall. In lines 1 and 2 we report the average effect after two and five years, respectively. Line 3 shows the placebo effect. Line 4 shows the mean of the outcome at baseline—i.e., one period before the event.

Furthermore, the event studies show that they fluctuate around zero similarly in pre- and post-event periods, suggesting no effect.

Three other points are worth mentioning regarding our results on mortality. First, these estimates are somewhat unstable to the inclusion of other controls. If we control for time trends based on some baseline characteristics, the estimates of the average effects for deaths of despair and alcohol-related deaths, for instance, change their signs. Second, if we look at a longer horizon (up to eight years after the event), the estimates converge to zero, which would not be expected if we had nonzero effects but lacked statistical power to detect them. Third, we also estimate the cumulative effects of the reform eight years after CAPS’ introduction and find no significant effects.

In sum, we find no evidence of an effect of CAPS on mental health mortality. We doubt that lack of statistical power plays a major role here, since we can detect reasonably small effects for most outcomes, especially our grouped measure of deaths of despair. Moreover, since most patients switching from inpatient care have a severe condition, we expect these effects to be larger than they would be if CAPS treated only people with milder conditions. Of course, mortality does not reflect mental health perfectly; hence, it is still possible that the centers affect mental health in other, less extreme ways.

E. Homicides

Figure 8 and Table 5 show the effects of CAPS on homicides. Estimates indicate that before CAPS’ introduction, treated and control municipalities had very similar

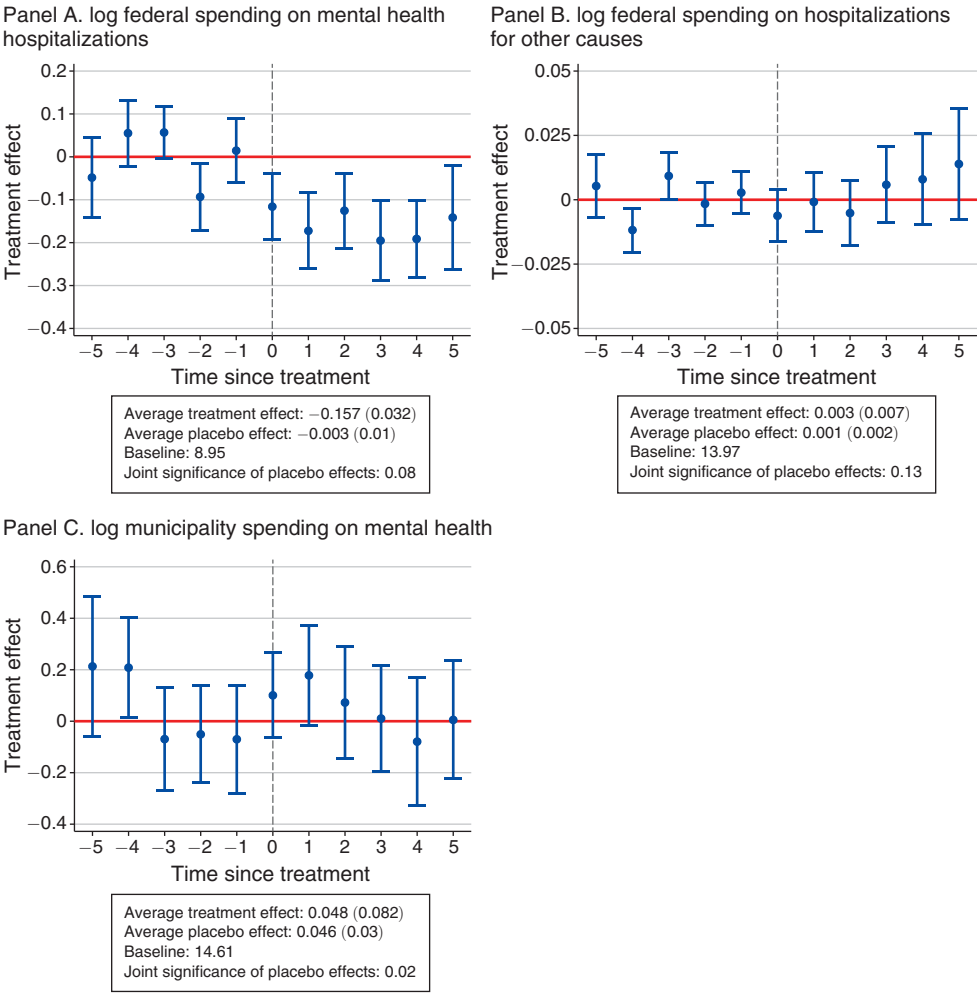


FIGURE 6. EFFECTS OF CAPS ON $\log(\text{PUBLIC HEALTH SPENDING})$

Notes: This figure plots 95 percent confidence intervals computed with a municipality-level clustered bootstrap and dynamic and placebo DID estimators for CAPS' effects on federal spending on mental health hospitalizations (panel A), federal spending on hospitalizations for other causes (panel B), and municipal mental health spending (panel C). All spending variables are in logs. Placebo DID estimators estimate the CAPS' effects had the treatment occurred in a placebo, pretreatment period. They use a varying baseline period—the one immediately before the placebo treatment date—so we do not normalize relative to a unique period. Controls include municipality GDP per capita, PBF spending per capita, a series of indicators for age-by-gender population bins, state $\times$ year fixed effects, and linear trends of pretreatment characteristics. Pretreatment municipality characteristics included are: Theil index, poverty rate, unemployment rate, illiteracy rate, share of rural population, log of population, log of social spending, number of mental health providers, number of mental health offices, municipality area, altitude, distance to capital, temperature, and rainfall. The Average Treatment Effect computes a weighted average of the dynamic estimators, giving to each estimator a weight proportional to the number of switchers the estimator applies to. In parentheses, standard errors are computed with a municipality-level clustered bootstrap. Baseline indicates the sample mean values for the treated in the first pre-CAPS period.

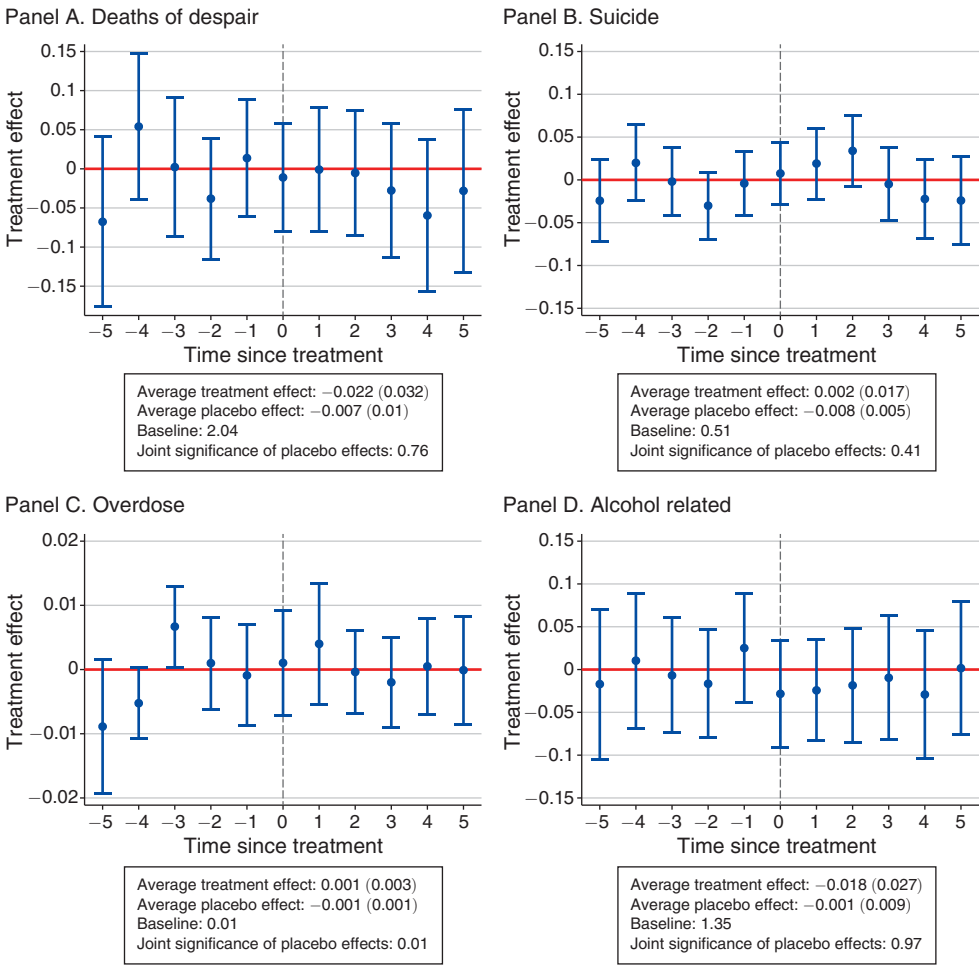


FIGURE 7. EFFECTS OF CAPS ON DEATHS OF DESPAIR: DEATHS PER 10,000 PEOPLE

Notes: This figure plots 95 percent confidence intervals computed with a municipality-level clustered bootstrap and dynamic and placebo DID estimators for the CAPS' effects on the number of deaths per 10,000 people associated with mental health. Panel A shows the results on deaths of despair, which include deaths by suicide (ICD-10 X60-X84), overdose (ICD-10 X40-X45, Y10-Y15, Y45, Y47, Y49), and alcohol-related deaths (which are a combination of alcoholic liver disease (ICD-10 K70, K73-K74) and deaths by mental disorders, which in the data are mainly associated with alcohol (ICD-10 F00-F99)). Panels B, C, and D show the results for each category separately. Placebo DID estimators estimate the CAPS' effects had the treatment occurred in a placebo, pretreatment period. They use a varying baseline period—the one immediately before the placebo treatment date—so we do not normalize relative to a unique period. Controls include municipality GDP per capita, PBF spending per capita, a series of indicators for age-by-gender population bins, state $\times$ year fixed effects, and linear trends of pretreatment characteristics. Pretreatment municipality characteristics included are: Theil index, poverty rate, unemployment rate, illiteracy rate, share of rural population, log of population, log of social spending, number of mental health providers, number of mental health offices, municipality area, altitude, distance to capital, temperature, and rainfall. The Average Treatment Effect computes a weighted average of the dynamic estimators, giving to each estimator a weight proportional to the number of switchers the estimator applies to. In parentheses, standard errors are computed with a municipality-level clustered bootstrap. Baseline indicates the sample mean values for the treated in the first pre-CAPS period.